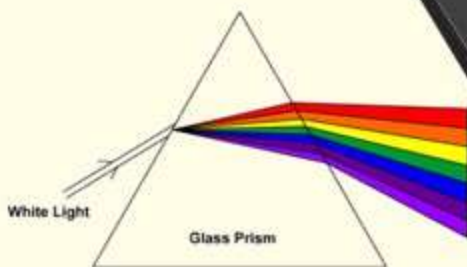
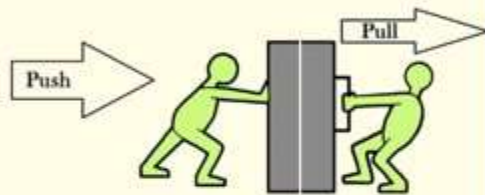


PHYSICS

FORCE – NEWTON'S LAWS



MODULE - 10



BALANCED FORCES

OBJECTS AT REST



STAY AT REST



OBJECTS IN MOTION



STAY IN MOTION



No acceleration in both cases





WHAT DO THESE PICTURES SUGGEST ?



Fruits fall off while
Shaking a tree



Dusting a carpet vigorously removes the
loose dust

A body tends to be in the state until an unbalanced force is acted upon



WHAT DO THESE PICTURES SUGGEST?



Sudden Brakes are applied



Sudden stopping of the giant swing

A body tends to be in the state until an unbalanced force is acted upon



NEWTON'S FIRST LAW

An object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.

BREAKUP OF THE LAW

- A body will resist motion when
- The body was at rest and is attempted to move
- The body was in motion and is attempted to stop
- The body was moving in one direction and it is attempted to change the direction



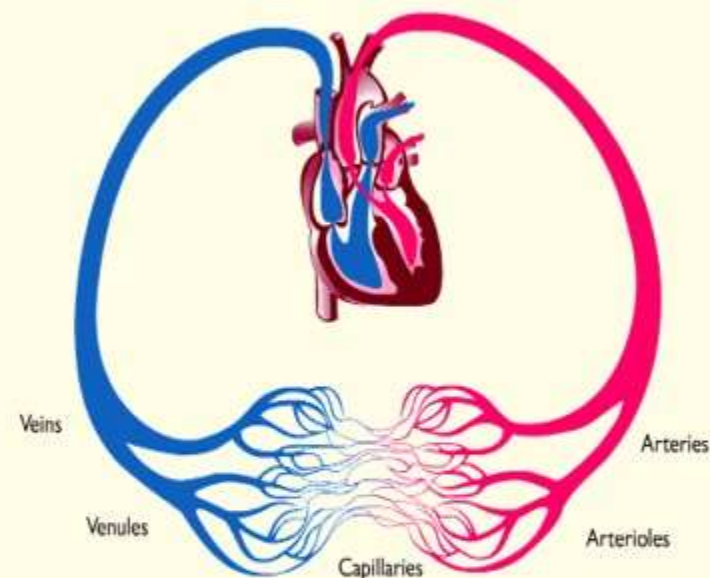
Inertia is the resistance of any physical object to any change in its state of motion



APPLICATIONS OF NEWTON'S FIRST LAW



Seat belts are highly useful to neutralize the inertia of motion.



Blood flows through the head to the feet quickly when descending the elevator



INERTIA AND MASS

*Which brick will offer least resistance of motion
when pushed slightly?*



Mortar brick



Styrofoam

More mass will offer more inertia

**QUESTION 1**

Suppose you are in a weightless environment. Would you require a force to set an object in motion?

- a) Force is required
- b) No force is required
- c) A slight force is required
- d) The question is invalid

Answer – Yes, force is required to move an object even in space.

Because, an object has mass even in space. So an object will have inertia. Thus Force is required to overcome the inertia.



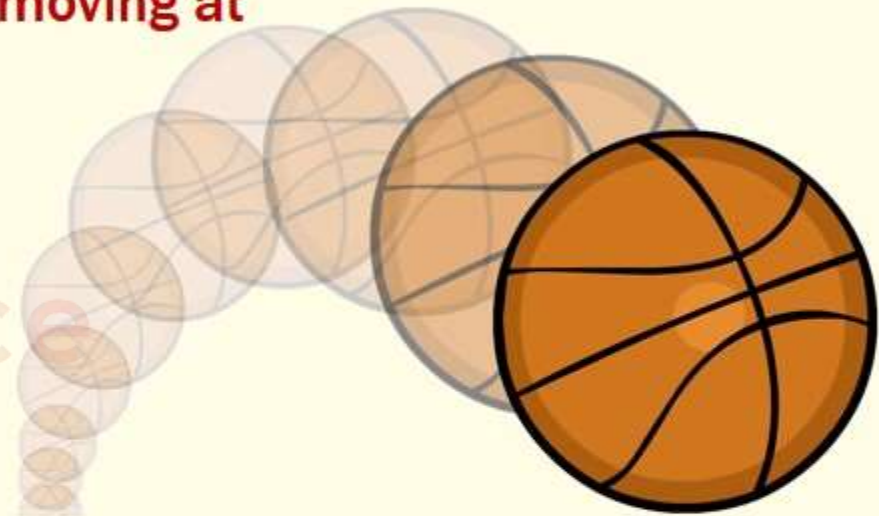
**QUESTION 2**

A 4-kg object is moving horizontally with a speed of 10 m/s. How much net force is required to keep the object moving at this speed and in this direction?

- a) -10N
- b) 0 N
- c) +5 N
- d) +10N

Answer – The net force required is 0 N

Because, when an object is moving with a constant speed, the object will continue to move unless acted upon by an unbalanced force.



UNBALANCED FORCES

**THERE IS AN
ACCELERATION**

**Acceleration
depends directly
on the net force**

**Acceleration depends
inversely on the
objects mass**





NEWTON'S SECOND LAW

The acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force, in the same direction as the net force, and inversely proportional to the mass of the object.

$$F = m \cdot a$$

$$1\text{N} = 1\text{kg} \cdot 1\text{ms}^{-2}$$

BREAKUP OF THE LAW

- Entire emphasis on the net force(vector sum of all forces).
- Net force equals mass times acceleration.
- The acceleration is in the same direction of the net force.
- The acceleration is inversely proportional to the mass of the object.

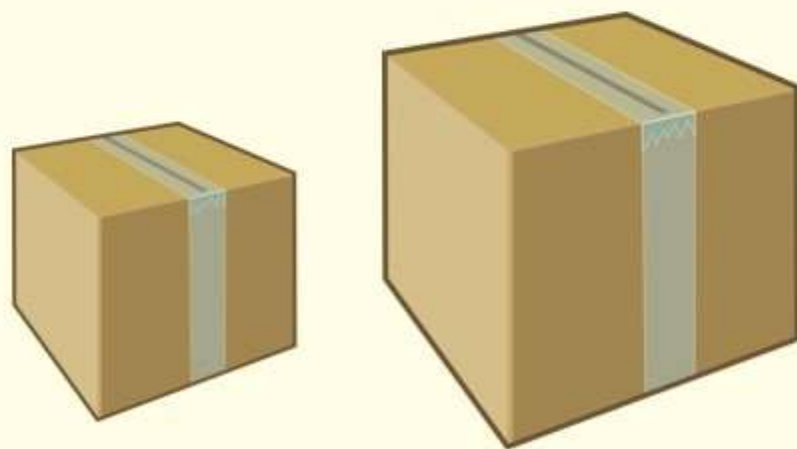


**QUESTION 3**

Determine the accelerations that result when a 12N net force is applied to a 3-kg object and then to a 6-kg object.

- a) 4ms^{-2} and 2ms^{-2}
- b) 2ms^{-2} and 4ms^{-2}
- c) 4ms^{-2} and 8ms^{-2}
- d) 8ms^{-2} and 2ms^{-2}

Hint: Use $F = m \cdot a$





NEWTON'S SECOND LAW – THE IMPACT OF FAST ACCELERATION



- A fielder pulls his hand gradually with the moving ball while holding a catch.
- This is because if the ball is stopped suddenly the velocity decreases to 0 in very short time.
- The acceleration of the stopping ball will be large, thus the impact will be large.
- To stop a fast moving ball, the acceleration should be brought down gradually to decrease the Force felt by the hands.

But why do the hands feel any force at all?





NEWTON'S THIRD LAW

For every action, there is an equal and opposite reaction

BREAKUP OF THE LAW

- There is a pair of forces in every interaction.
- The pair of forces acts on two different bodies.
- The size of the forces on the first object equals the size of the forces in the second object.
- The direction of the force of the first object is opposite to the direction of the force of the second object.
- Force always come in pairs – action and reaction forces.

Equal forces do not always have equal effects



EXAMPLES OF NEWTON'S THIRD LAW

FLYING MOTION OF BIRDS

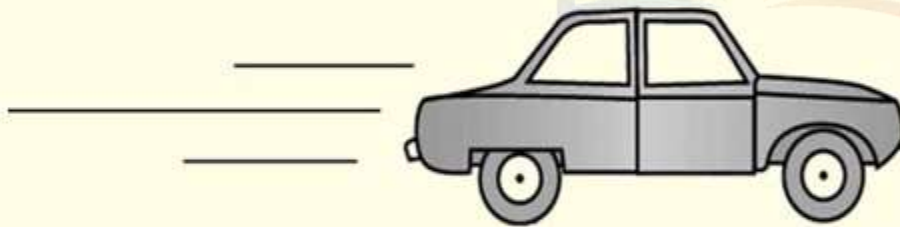


- A bird flies by use of its wings.
- The wings of a bird push air downwards.
- Since forces result from mutual interactions, the air must also be pushing the bird upwards.
- The size of the force on the air equals the size of the force on the bird;
- the direction of the force on the air (downwards) is opposite the direction of the force on the bird (upwards).
- For every action, there is an equal (in size) and opposite (in direction) reaction.
- Action-reaction force pairs make it possible for birds to fly.



EXAMPLES OF NEWTON'S THIRD LAW

MOTION OF THE CAR



- A car is equipped with wheels that spin.
- As the wheels spin, they grip the road and push the road backwards.
- Since forces result from mutual interactions, the road must also be pushing the wheels forward.
- The size of the force on the road equals the size of the force on the wheels (or car);
- the direction of the force on the road (backwards) is opposite the direction of the force on the wheels (forwards).
- For every action, there is an equal (in size) and opposite (in direction) reaction.
- Action-reaction force pairs make it possible for cars to move along a roadway surface.



**QUESTION 5**

The recoil of gun is the result of action-reaction force pairs. A gunpowder explosion creates hot gases that expand outward allowing the rifle to push forward on the bullet. Consistent with Newton's third law of motion, the bullet pushes backwards upon the rifle. The acceleration of the recoiling rifle is ...

- a) Greater than the acceleration of the bullet.
- b) Smaller than the acceleration of the bullet.
- c) The same size as the acceleration of the bullet.
- d) None of these



Answer – The force on the rifle equals the force on the bullet. Yet, acceleration depends on both force and mass. The bullet has a greater acceleration due to the fact that it has a smaller mass.

Remember: acceleration and mass are inversely proportional



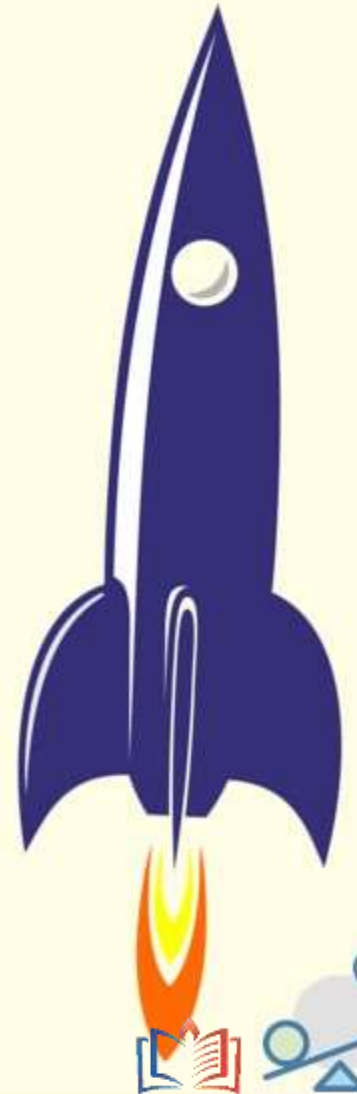
QUESTION 6

Space travel was believed to be impossible because there was nothing that rockets could push off of in space in order to provide the propulsion necessary to accelerate. This inability of a rocket to provide propulsion is because ...

- a) Space is void of air so the rockets have nothing to push off of.
- b) Gravity is absent in space.
- c) Space is void of air and so there is no air resistance in space.
- d) Rockets do accelerate in space and have been able to do so for a long time.

Answer - The fact is that rockets do accelerate. There is indeed nothing for rockets to push off in space - But that's no problem for rockets.

Rockets are able to accelerate due to the fact that they burn fuel and push the exhaust gases in a direction opposite the direction which they wish to accelerate.



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